(e.g. the bottom landing of a stairway or complex areas of the structure withi n the space) and this should be taken into account.

7.3 All ships should carry at least two sets of gas detection equipment as required by SOLAS regulation XI -1/7, taking into account the Guidelines to facilitate the selection of

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SOURCE: MSC.581(110).pdf
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SOLAS regulation XI -1/7, taking into account the Guidelines to facilitate the selection of portable atmosphere testing instruments for enclosed spaces as required by SOLAS regulation XI-1/7 (MSC.1/Circ.1477). Any ship which may carry a cargo capable of generating hazardous vapours and which requires regular entry into the cargo space for cleaning or inspection should carry two sets of gas detection equipment in addition to those required by SOLAS regulation XI-1/7 for assessing the risk to personnel entering the space. The use of flexible hoses or fixed sampling lines, which reach remote areas within the enclosed space, may allow for safe testing without the need to enter the space.

7.4 After completion of a suitable risk assessment targeted to the space to be entered steady readings of all of the following should be obtained:

.1 20.9% oxygen by volume;

.2 the level of carbon dioxide has been checked and is less than 0.5% by volume (5,000 ppm);

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.1 20.9% oxygen by volume;
.2 the level of carbon dioxide has been checked and is less than 0.5% by volume (5,000 ppm);

Note: National requirements may differ when determining the safe atmosphere range for gases stated above.

.3 less than 1% of lower flammable limit (LFL) on a suitably sensitive combustible gas indicator, where the assessment has determined that there is potential for flammable gases or vapours; and

.4 less than 50% of the occupational exposure limit (OEL) of any toxic vapours and gases.

7.5 If these conditions cannot be met, additional ventilation should be applied to the space and retesting should be conducted after a suitable interval. Entry should be allowed only after all the above conditions are met. When the atmosphere remains, or is suspected to be unsafe, then the guidance contained within section 9 should apply.

RESOLUTION MSC.581(110) (adopted on 27 June 2025)
REVISED RECOMMENDATIONS FOR ENTERING
ENCLOSED SPACES ABOARD SHIPS

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RESOLUTION MSC.581(110) (adopted on 27 June 2025)
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7.6 Any gas testing should be carried out with ventilation to the enclosed space stopped, and after conditions have stabilized, in order to obtain accurate readings.
7.7 Where the assessment has determined that there is potential for the presence of toxic gases and vapours, appropriate testing should be carried out, using fixed or portable gas or vapour detection and/or monitoring equipment. The readings obtained by this equipment should be below the occupational exposure limits for the toxic gases or vapours given in accepted national or international standards, in accordance with paragraph 7.4. It should be noted that testing for flammability or oxygen content does no t provide a suitable means of measuring for toxicity, nor vice versa.

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noted that testing for flammability or oxygen content does no t provide a suitable means of measuring for toxicity, nor vice versa.
7.8 It should be emphasized that the internal structure of the space, cargo, cargo residues and tank coatings may also present situations where oxygen -deficient areas may exist, and should always be suspected, even when an enclosed space has been satisfactori ly tested as being suitable for entry. This is particularly the case for spaces where the path of the supply and outlet ventilation is obstructed by structural members or cargo.
8 PRECAUTIONS DURING ENTRY
8.1 The atmosphere should be tested frequently whilst the space is occupied, and persons should be instructed to leave the space should there be any sign of deterioration in the conditions.
8.2 Persons entering enclosed spaces should be provided with calibrated and tested personal gas detection instrument or instruments that monitor the levels of oxygen, carbon
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personal gas detection instrument or instruments that monitor the levels of oxygen, carbon dioxide, flammable gases or vapours, toxic gases (including carbon monoxide), and any other gases identified in the risk assessment.
8.3 Ventilation should be maintained while the space is occupied. After a break and before re-entry, the atmosphere must be retested, the results verified as being acceptable then recorded. In the event of failure of the ventilation system, any persons in the space should leave immediately.
8.4 Particular care should be taken when working on piping and valves within the space which may contain hazardous gases, vapours or liquids. If conditions change during the work, increased frequency of testing of the atmosphere should be performed. Examples of changing conditions are as follows increase or decrease in ambient temperature, the need to use oxygen fuel torches or other welding equipment, the need to use a mobile plant, other activities
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oxygen fuel torches or other welding equipment, the need to use a mobile plant, other activities in the space that may involve development of vap our suc h as cleaning debris, collecting sediments, painting within the enclosed spaces, changes in the ship's trim and list while working within the enclosed space. Conditions can also deteriorate during work breaks

because of changes to the ventilation, the dist urbance of bilge contents, sludge or stagnant water, or the leakage of external contaminants into the space. Depending on the result of the testing, a decision should be made as to whether it is safe to continue working.

8.5 In the event of an emergency, under no circumstances should the attendant enter the space before help has arrived and the situation has been evaluated to ensure the safety of those entering the space to undertake rescue operations. Only properly trained and equipped personnel should perform rescue operations in enclosed spaces.

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personnel should perform rescue operations in enclosed spaces.

9 ADDITIONAL PRECAUTIONS FOR ENTRY INTO A SPACE WHERE THE ATMOSPHERE IS KNOWN OR SUSPECTED TO BE UNSAFE

9.1 In preparing to enter an enclosed space, every effort should be made to ensure that it is safe to do so in order to undertake the activity required within the space. Entry into enclosed spaces where the atmosphere is known or suspected to be unsafe requires very careful consideration, including an assessment of the hazards, residual risks and mitigations that need to be undertaken. Spaces that have not been tested should be considered unsafe for persons to enter.

9.2 When considering the planned work activity within an enclosed space where the

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9.2 When considering the planned work activity within an enclosed space where the atmosphere is known or suspected to be unsafe, suitably designed and constructed breathing apparatus of positive pressure type should always be worn, and only personnel trained i n its use should be allowed to enter the space. Air -purifying respirators, dust masks and canister face masks do not provide a supply of clean air from a source independent of the atmosphere within the space and should not be used. Emergency Escape Breathing Devices (EEBDs) are not suitable to use for entry into enclosed spaces.

9.3 Notwithstanding paragraph 6.5.6, persons entering enclosed spaces that may contain a suspected atmospheric hazard should be provided with calibrated and tested personal gas detector suitable for the gas or gases assessed as likely to occur in the space.

9.4 Rescue harnesses should be worn and unless impractical, lifelines should also be

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9.4 Rescue harnesses should be worn and unless impractical, lifelines should also be used. A means to facilitate evacuation from the enclosed space should be available and ready for use, as per emergency response plan.

9.5 Appropriate protective clothing should be worn, particularly where there is any risk of toxic substances or chemicals coming into contact with the skin of those entering the space.

9.6 The advice in section 11 concerning actions to be taken in an emergency is

particularly relevant in this context.
10 HAZARDS RELATED TO SPECIFIC TYPES OF SHIPS OR CARGO
10.1 Dangerous goods in packaged form
10.1.1 The atmosphere of any space containing dangerous goods may put at risk the health or life of any person making an entry. Dangers may include flammable, toxic, corrosive or asphyxiant gases or vapours, residues on packages and spilled material. The same haz ards may be present in spaces adjacent or connected to the cargo spaces. Information on the
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may be present in spaces adjacent or connected to the cargo spaces. Information on the hazards of specific substances is contained in the IMDG Code, the SDS and the Shippers Declaration. If there is evidence or suspicion that leakage of dangerous sub stances has occurred, the precautions specified in these recommendations should be followed.
10.1.2 Personnel required to deal with spillages or to remove defective or damaged packages should be appropriately trained and wear suitable breathing apparatus and protective clothing commensurate with the task.
10.2 Liquid bulk
The industry has produced extensive advice to managers, operators and crews of ships engaged in the bulk carriage of oil, chemicals and liquefied gases, in the form of specialist safety guides. Information contained in the guides on enclosed space entry am plifies these recommendations. In particular, for chemical tankers as defined in SOLAS regulation VII/8, the
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recommendations. In particular, for chemical tankers as defined in SOLAS regulation VII/8, the diversity of bulk liquid chemicals carried in some cargo tanks and the limitations in vapour detection technologies may pose complications that require specific and targeted mitigation.

10.2.1 Safety information
Safety information2 for the correct handling and carriage of liquid bulk and gas cargoes is set out in the SDS or other cargo information provided by the shipper to the master. Such information should be made available to all ship and shore personnel involved in the handling of the cargo, cargo slops and tank cleaning.
10.2.2 Use of nitrogen as an inert gas3
Nitrogen is a colourless and odourless gas that, when used as an inert gas, causes oxygen